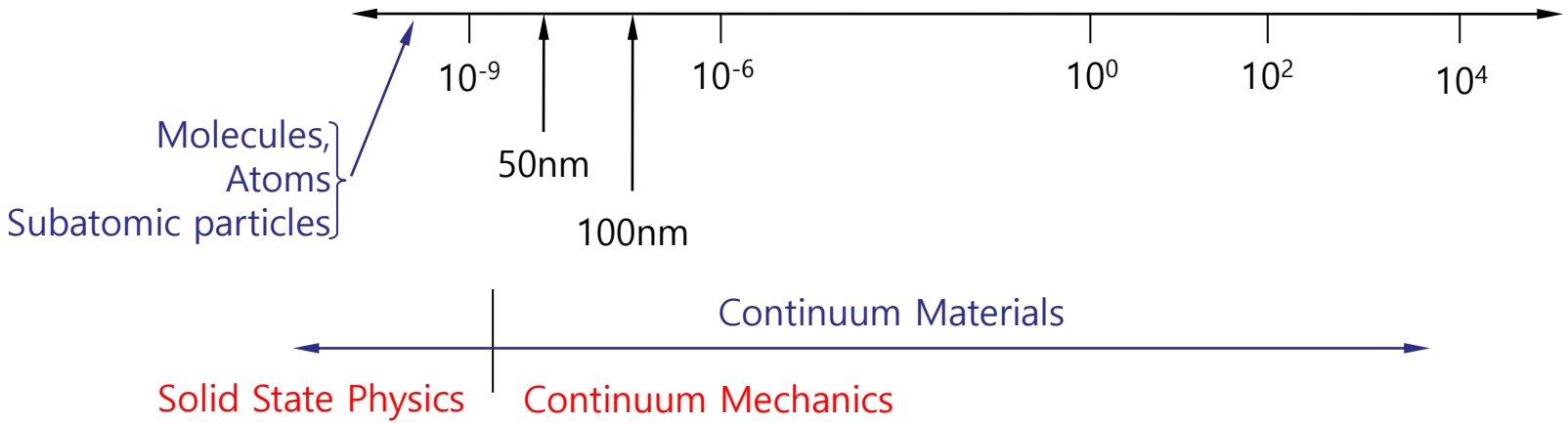
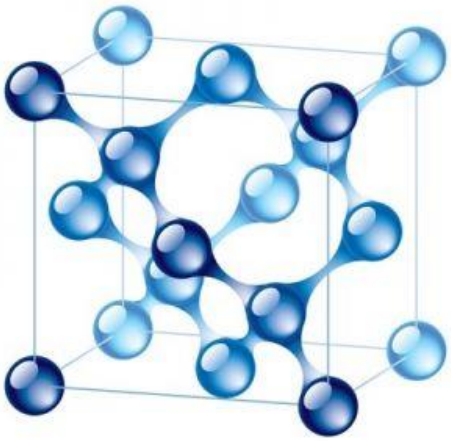
Chap. 1 Background

1.1 Introduction

▼ Scale of materials (m)

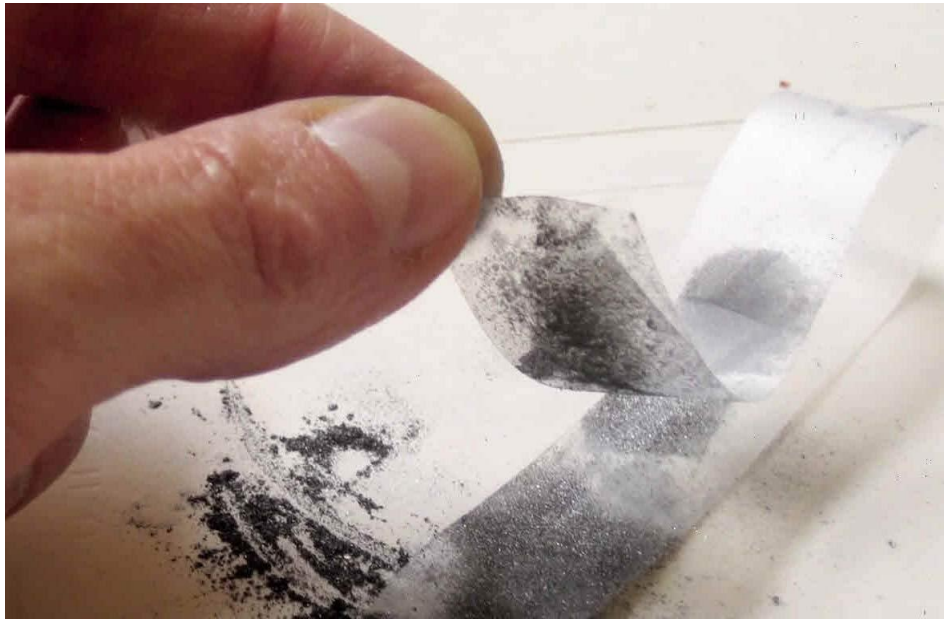


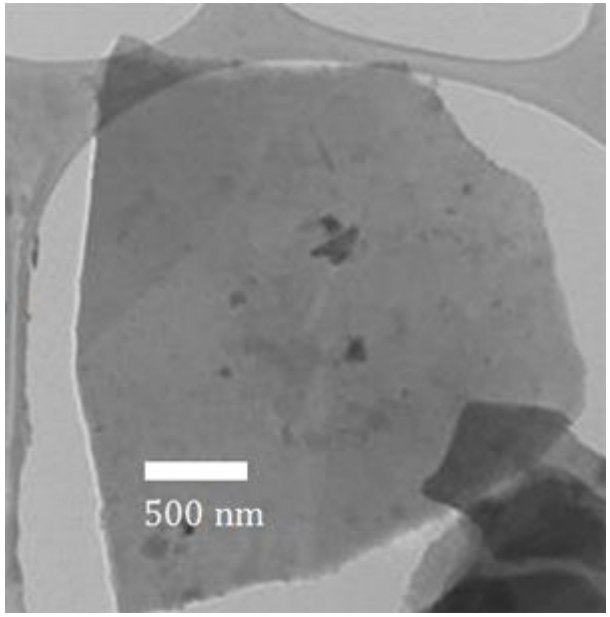
=> We assume materials are homogeneous and continuously distributed over their volumes.



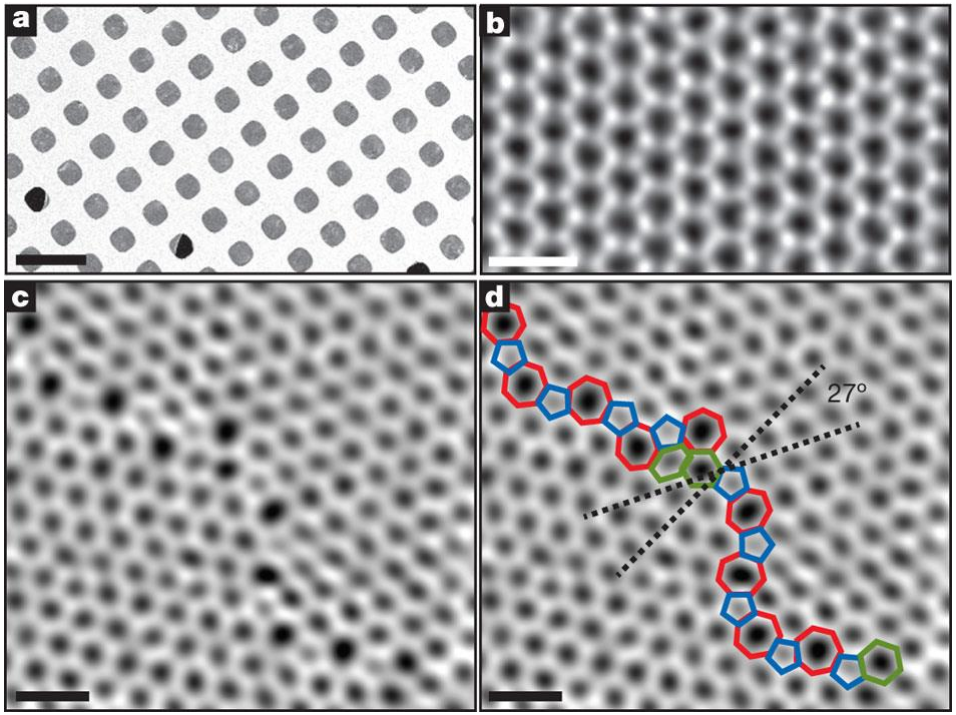
Atomic structure of Diamond

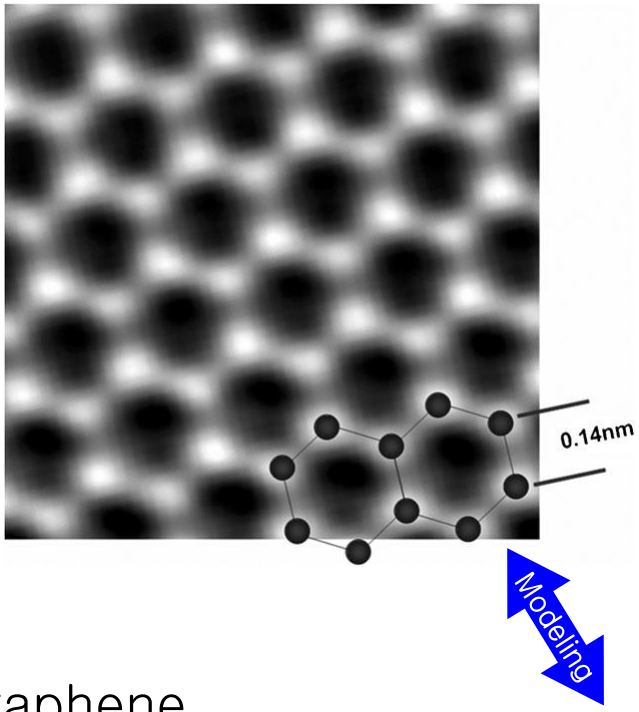
Diamond



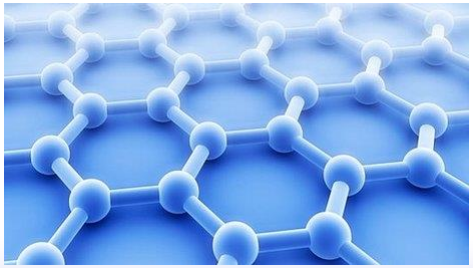


Graphene

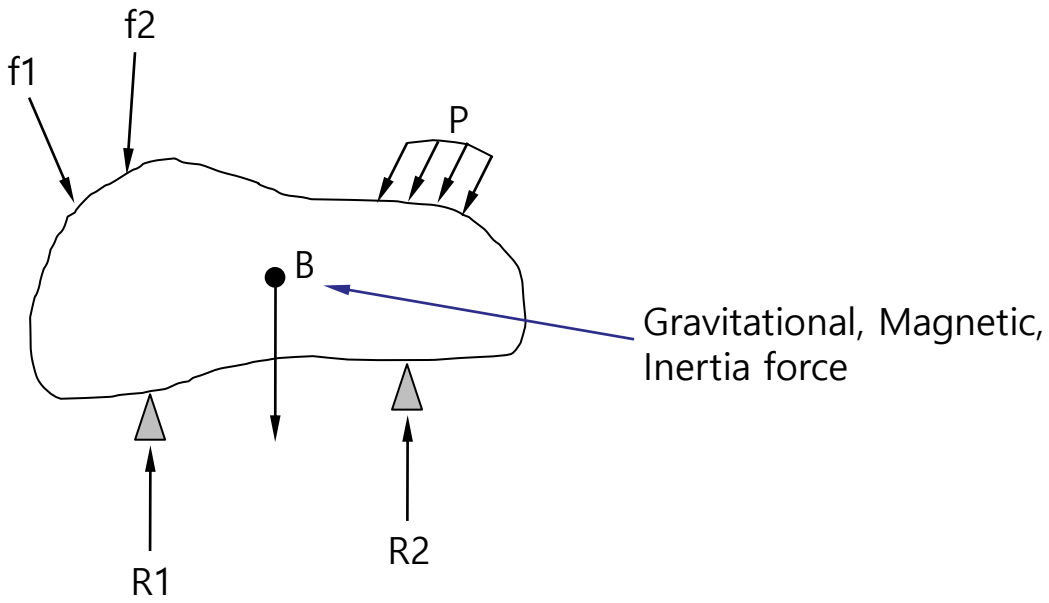




Atomic structure of graphene



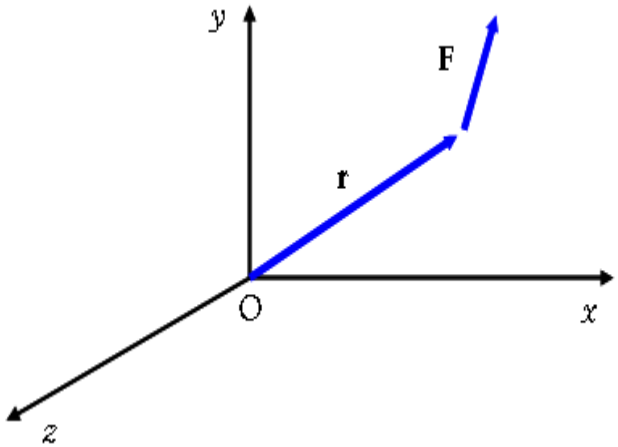
=> The mechanics of materials is concerned with relationships of external effect (forces & moments) to internal stresses & strains.
(internal forces & deformation)

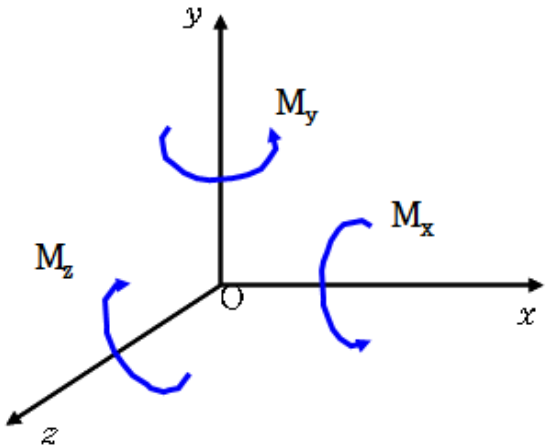


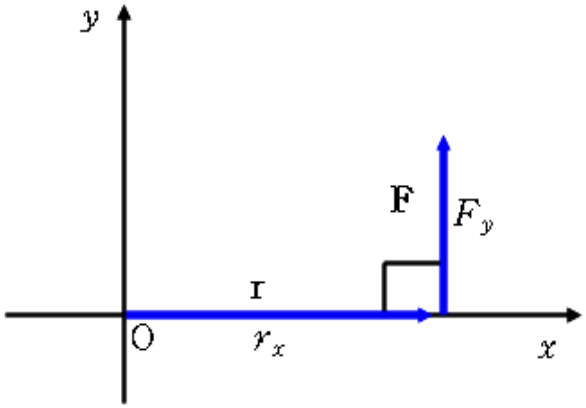
=> Internal forces are the forces that hold together the particles forming the body.

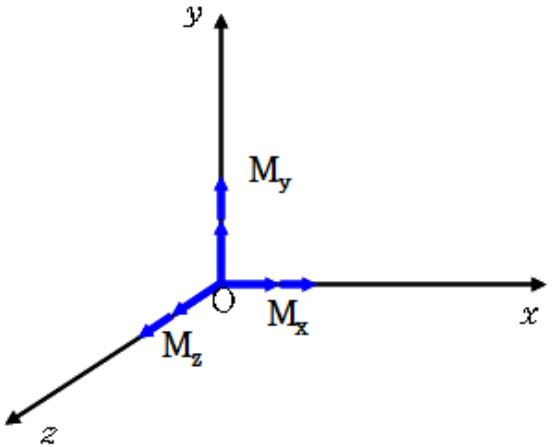
▼ In this text, body forces are neglected & forces are applied steadily and slowly (=static loading).

▼ Definition of force & moment

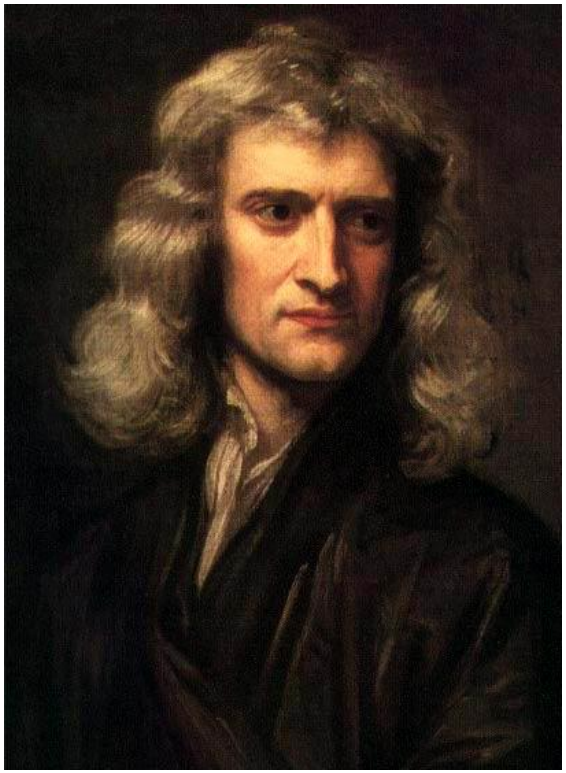




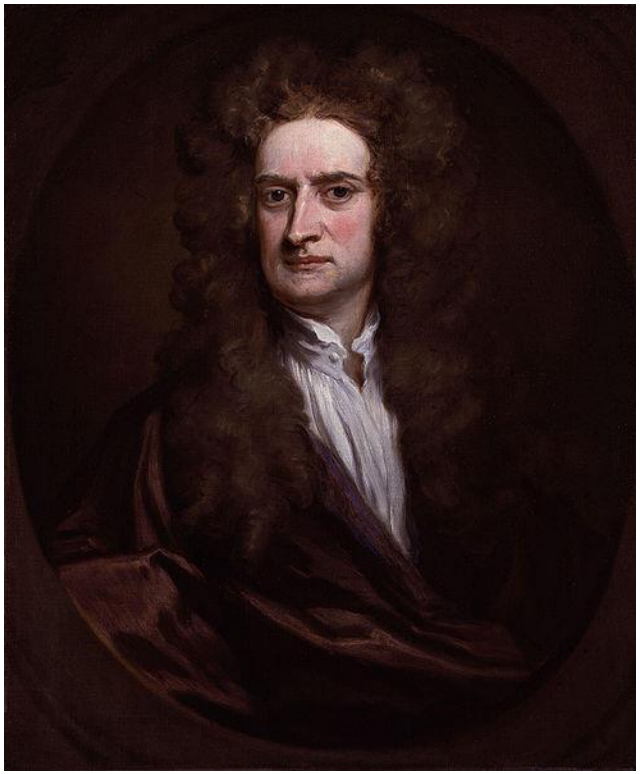




Isaac Newton (1642-1727)

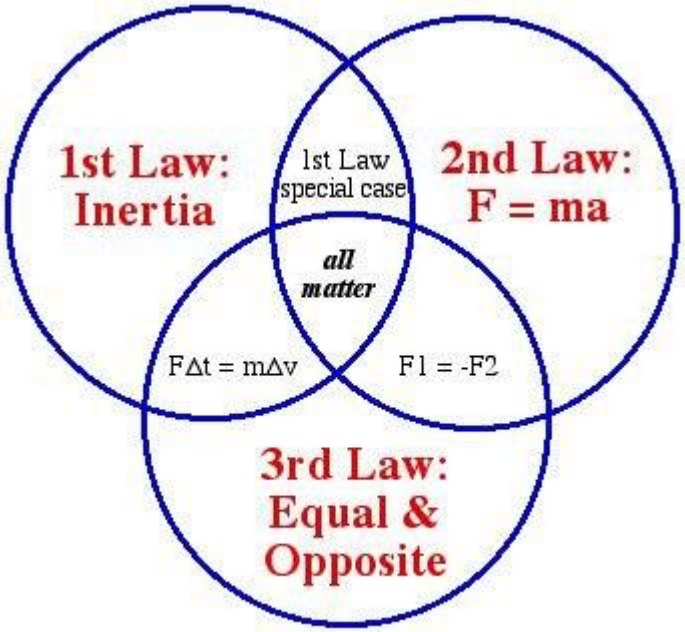


1689년



1702년

- **Newton의 운동법칙**
  - Inertia
  - Acceleration
  - Action & Reaction



Newton's Laws

- **Newton의 운동법칙**
  - Inertia

    $\Rightarrow \quad 0 = 0$

  - Acceleration

    $\Rightarrow \quad \sum F = ma$

  - Action & Reaction

    $\Rightarrow \quad F_1 = F_2$

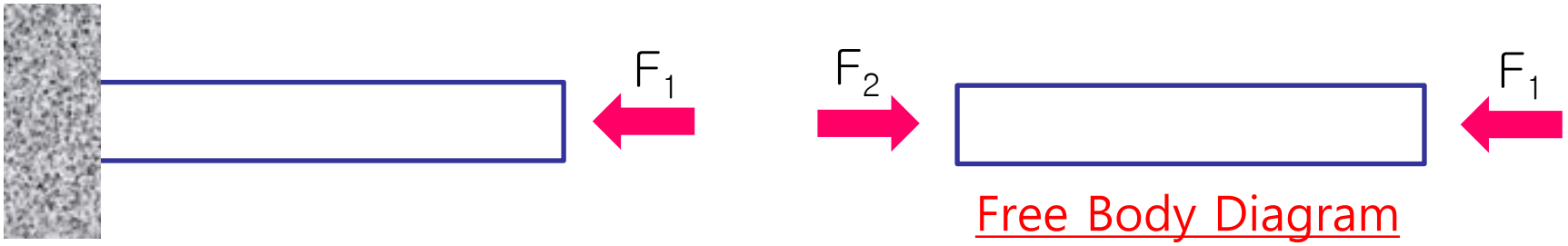
▼ Equilibrium state (Newton's 1st & 2nd law)

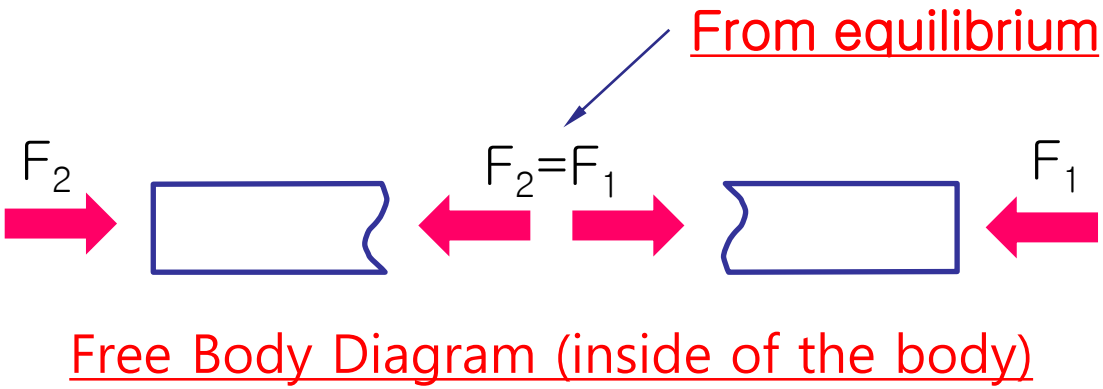
$\sum F = ma$: Dynamic problem

$\sum F = 0$: Static problem

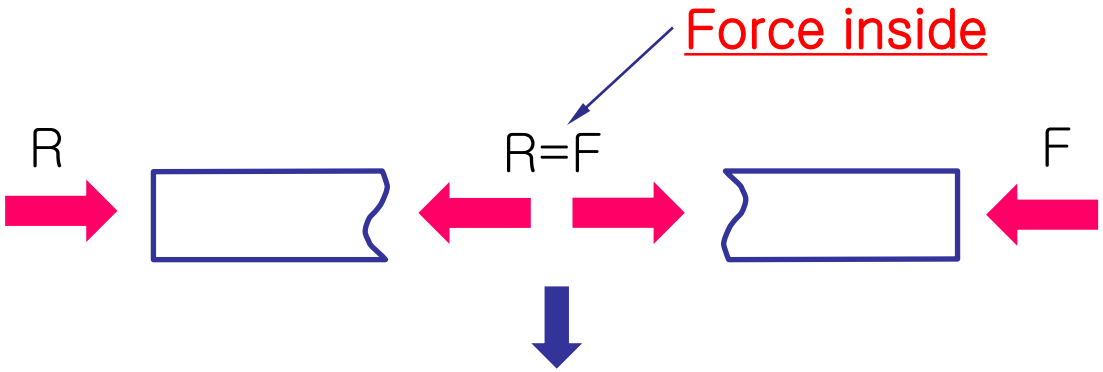
Equilibrium : R=F

▼ Free body diagram (Newton's 3rd law)





▼ Stress & strain



STRESS=F/A

STRAIN (From deformation)

1.2 Scope of treatment

1. Analysis of the stress and deformations within a body subject to a prescribed system of forces (▼ solving the governing equations).

2. Determination of the limiting (allowable) values of load by theoretical analysis or by experiment.

3. Determination of the body shape and selection of the materials.
   => Design concept, more particularly, optimum design.